

Youcheng Zong

PhD Student in Control Science and Engineering

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Summary

- PhD student at Northeastern University (China), working on industrial LLMs, intelligent metallurgy, process industry intelligence, and industrial time-series analysis for complex industrial processes.

Education

PhD student, Control Science and Engineering	Northeastern University, College of Information Science and Engineering	Shenyang, China Sept 2024 – present
Bachelor of Engineering in Automation	Anhui University of Technology, School of Electrical and Information Engineering	Ma'anshan, Anhui, China Sept 2017 – June 2021

Academic Interests

Industrial AI: Industrial LLMs, human-AI collaborative decision support, process industry intelligence

Metallurgical Process Intelligence: Intelligent metallurgy, blast furnace state recognition, continuous casting analysis

Industrial Time Series: Temperature trend prediction, predictive modeling, data-driven process monitoring

Publications

LLM-Driven Human-AI Collaborative Decision Support System for Complex Industrial Processes: A Case Study in Metallurgy	Apr 2026
First-author journal article on LLM-driven decision support for complex industrial processes. Youcheng Zong, Shuyue Feng, Zhenhe Ma, Jinhui Wang, Jianhua Liu, Chao Yang 10.1016/j.neunet.2026.109055 (Neural Networks)	
A meta-contrastive learning hybrid model for adaptive temperature trend prediction in variable ladle preheating	Dec 2025
First-author journal article on adaptive prediction for ladle preheating. Youcheng Zong, Jinhui Wang, Chunhua Yang, Chao Yang 10.1016/j.engappai.2025.112750 (Engineering Applications of Artificial Intelligence)	

Hybrid Grid Search and Bayesian optimization-based random forest regression for predicting material compression pressure in manufacturing processes

Feb 2025

First-author journal article on pressure prediction in manufacturing processes.

Youcheng Zong, Jinhui Wang, Chao Yang, Haibo Lin, Zhuchao Xia, Yi Luo

[10.1016/j.engappai.2024.109580](https://doi.org/10.1016/j.engappai.2024.109580) (Engineering Applications of Artificial Intelligence)

Iron-Tapping State Recognition of Blast Furnace Based on Bi-GRU Composite Model and Post-Processing Classifier

Sept 2023

First-author journal article on blast furnace iron-tapping state recognition.

Youcheng Zong, Jinhui Wang, Yuanyuan Wang, Su Xu, Guanyu Zheng

[10.1109/JSEN.2023.3300123](https://doi.org/10.1109/JSEN.2023.3300123) (IEEE Sensors Journal)

Academic Profiles

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